Using the RED DGVM equilibrium to infer demography from remote sensing

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The RED model

- RED simulates the size-structure of forests through updating the PFT number density (n) across plant mass (m) via the growth-rate g and mortality (γ):

$$\frac{\partial}{\partial t} n(m,t) - \frac{\partial}{\partial m}\left[n(m,t) \cdot g(m,t)\right] = -\gamma(m,t) \cdot n(m,t)$$

- PFT competition acts on the number of seedlings entering the size-structure. The coverage (ν) shades the seed flux entering n_0:

$$n_0 g_0 = \frac{\alpha P}{m_0}(1-\nu)$$

Where P is the net carbon assimilate across the grid-box, and α is the fraction of assimilate devoted to reproduction. (Argles, et al., 2020)

RED allometry

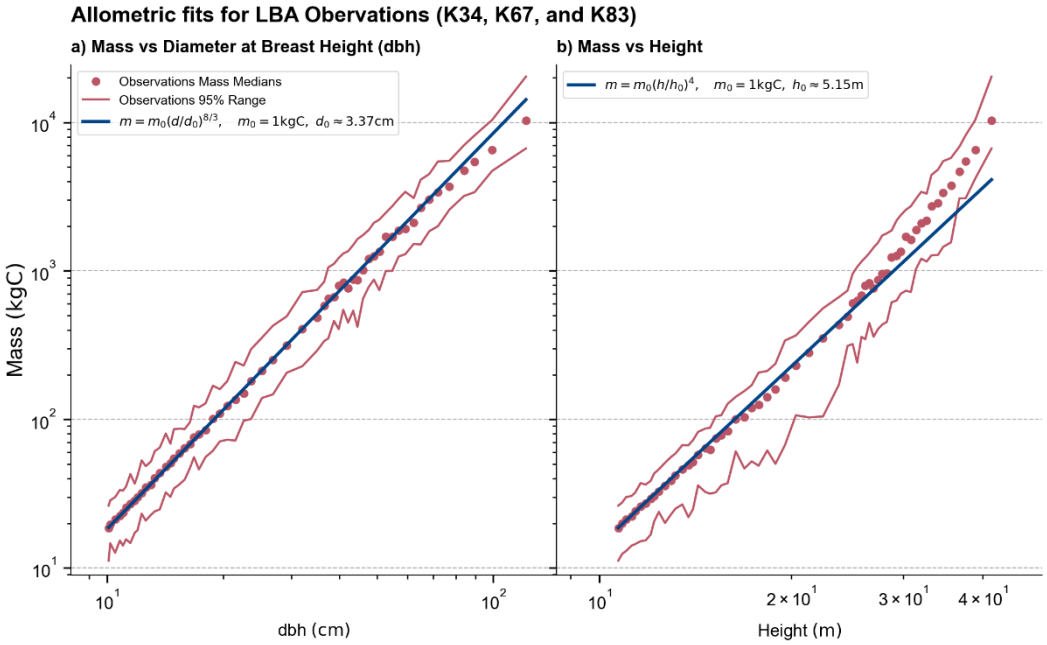
- Allometric scaling are good approximations for how tree physiology varies across tree mass.
- For example, for tree mass-diameter, the basic form of this relationship is:

$$d = d_0 \left(\frac{m}{m_0}\right)^{\frac{8}{3}}$$

Or more generally for arbitrary physical variable (y):

$$y = y_0 \left(\frac{m}{m_0}\right)^{\phi_y}$$

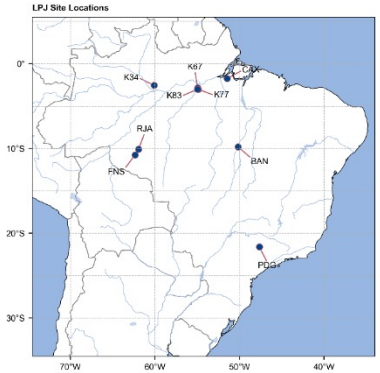
Data Sources: Wofsy, et al., 2008 (K34), Menton, et al., 2011 (K83), Chambers, et al., 2013 (K67), and Zanne, A. E., 2009 (DRYAD).



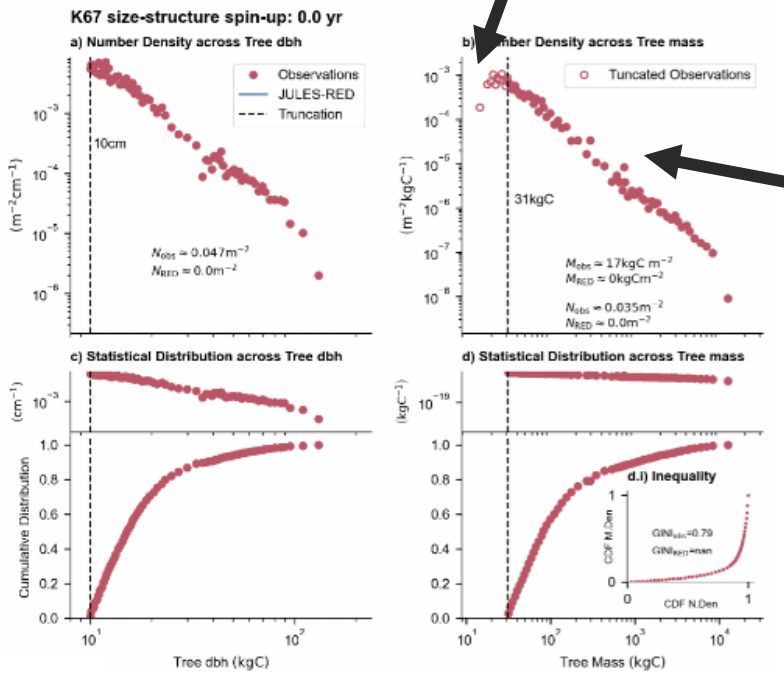
RED in motion

$$n_0 g_0 = \frac{\alpha P}{m_0}(1-\nu)$$

$$\frac{\partial}{\partial t} n - \frac{\partial}{\partial m}[ng] = -\gamma\, n$$



Shows the geographic locations of the LBA sites across Brazil, including the shorthand codes for each site (see Saleska, et al., 2013).

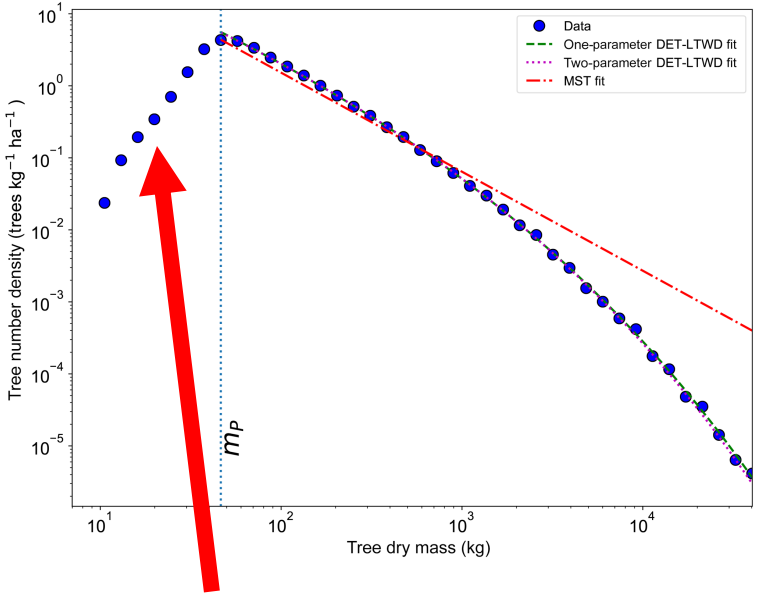


Introducing Demographic Equilibrium Theory (DET)

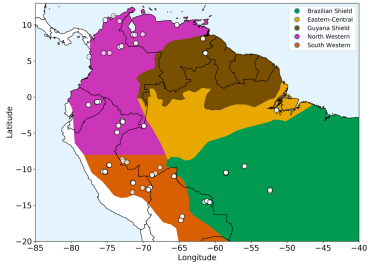
$$\frac{\partial}{\partial t} n(m,t) + \frac{\partial}{\partial m}\left[\left(\frac{m}{m_0}\right)^{\frac{3}{4}} g_0 n(m,t)\right] = \mu_0 n(m,t)$$

- Equation (Left Truncated Weibull Distribution, LTWD) describes the forest size structure if:
  1. Forest is in equilibrium $\frac{\partial}{\partial t} \to 0,\ \gamma(m) = n(t,m) = n(t+\Delta t, m)$
  2. Tree growth conforms to Metabolic Scaling Theory $g(m) = g_0 \left(\frac{m}{m_0}\right)^{\frac{3}{4}}$
  3. If tree mortality is constant through time
- To reduce the parameters we introduce a turnover term $\mu_0 = \gamma m_0 / g_0$, ($[\mathrm{kg}][\mathrm{yr}^{-1}]/[\mathrm{kg\ yr}^{-1}]$: dimensionless).

Evaluation of DET



$$n(m) = n_0 \left(\frac{m}{m_0}\right)^{-\frac{3}{4}} \exp\left\{4\mu_0 \left[1 - \left(\frac{m}{m_0}\right)^{\frac{1}{4}}\right]\right\}$$



- We have evaluated DET using the RAINFOR dataset (124 sites, https://www.forestplots.net).
- Metabolic scaling theory (MST), does not fully explain the distribution of larger trees.

Observations of (dbh ~ 10cm) do not capture all tree masses.

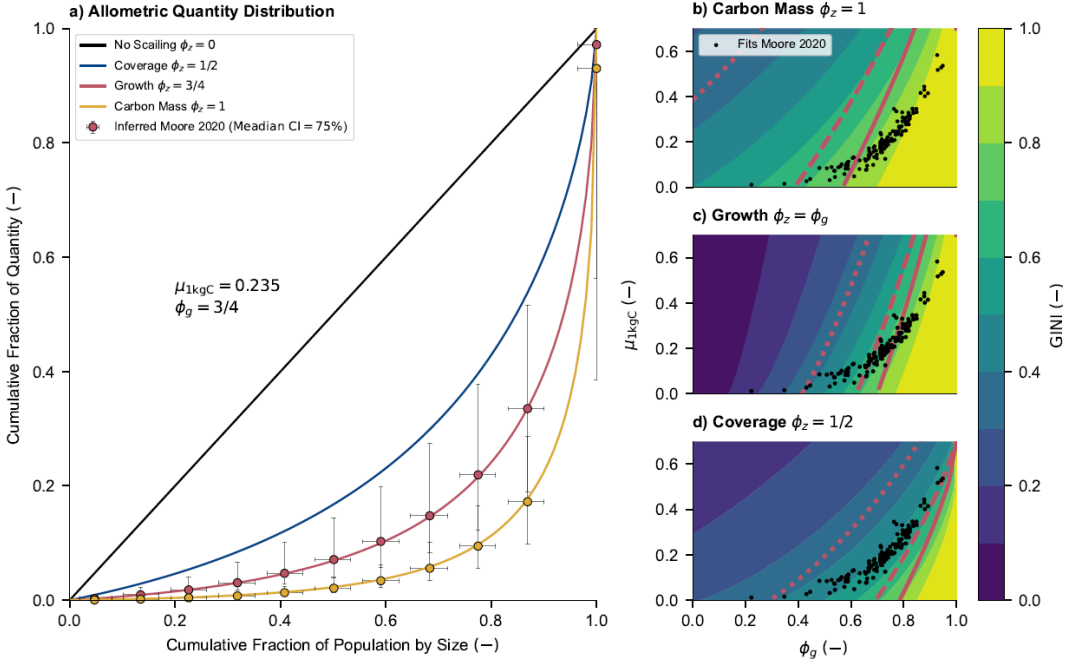
Moore, Jonathan R., et al. "Validation of demographic equilibrium theory against tree-size distributions and biomass density in Amazonia." Biogeosciences 17.4 (2020): 1013-1032.

Total Forest Equilibrium Attributes (Allometry + DET)

- Forest Total M

- Forest Bioma

$$M = m_0 N \left(1 \right.$$



$/4, y = g)$:

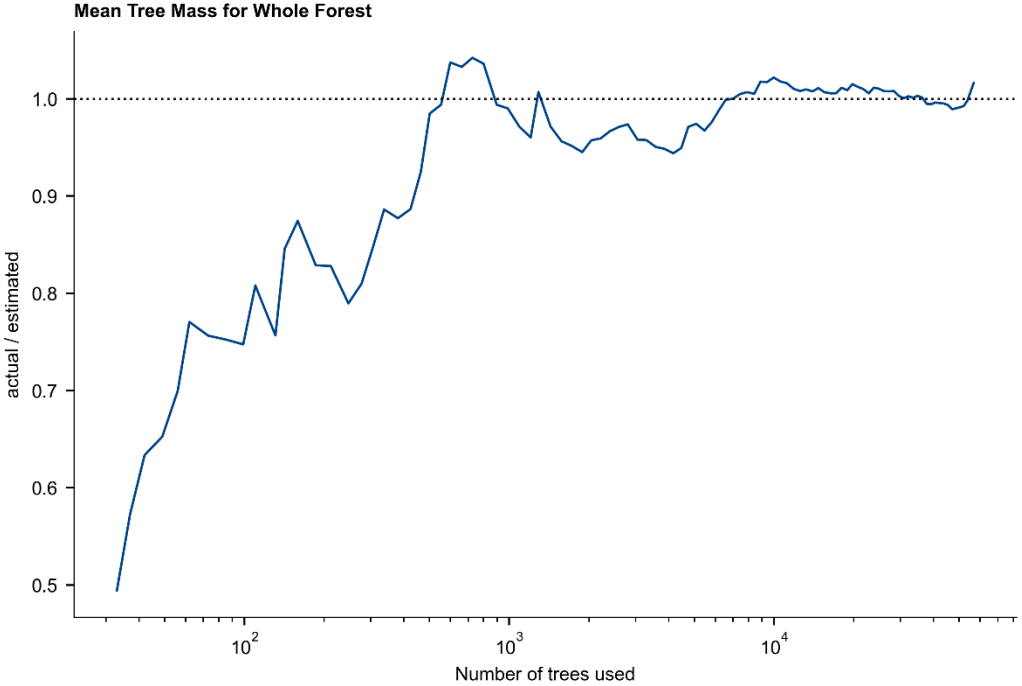
$\frac{}{\mu_0^2} + \frac{3}{32\mu_0^3}\Big)$

projected crown

$+ \frac{1}{8\mu_0^2}\Big)$

Using DET to infer Forest Quantities

- Another way to look at this. We only need to sample 1,000 trees out of 80,000 from the RAINFOR dataset to be reasonably close for the mean mass/
- For example, surveying all trees above 2,000 kgC can give reasonable estimates of the forest in total.
- By taking largest few trees as a sample, we can use DET to reproduce the lower distribution.



Bringing it back to RED or “Closed Form – DET”

- As a reminder, RED has the competitive constraint on the vegetation dynamics for the seedling flux:

$$n_0 g_0 = \frac{\alpha P}{m_0}(1-\nu)$$

- When we combine all these relationships, we end up with a two parameter equation for total forest coverage:

$$\nu = 1 - \left(\frac{n_0 g_0}{\alpha}\right)\frac{1}{1+\frac{3}{4\mu_0}+\frac{3}{8\mu_0^2}+\frac{3}{32\mu_0^3}}$$

- If we introduce these DET equilibrium solutions:

$$N = \frac{n_0 m_0}{\mu_0} = \frac{n_0 g_0}{\gamma}, \qquad G = g_0 N \qquad M = m_0 N$$

- In RED, α, is a PFT specific parameter held constant, and μ_0 is the ratio of mortality to growth.

or

- We can expand this relationship to get estimates for the total carbon density:

$$n_0 g_0 = N\left(\frac{\alpha}{\alpha_0}\right)\frac{G}{m_0}\left(\frac{1}{1+\frac{1}{2\mu_0}+\frac{1}{8\mu_0^2}}\right)$$

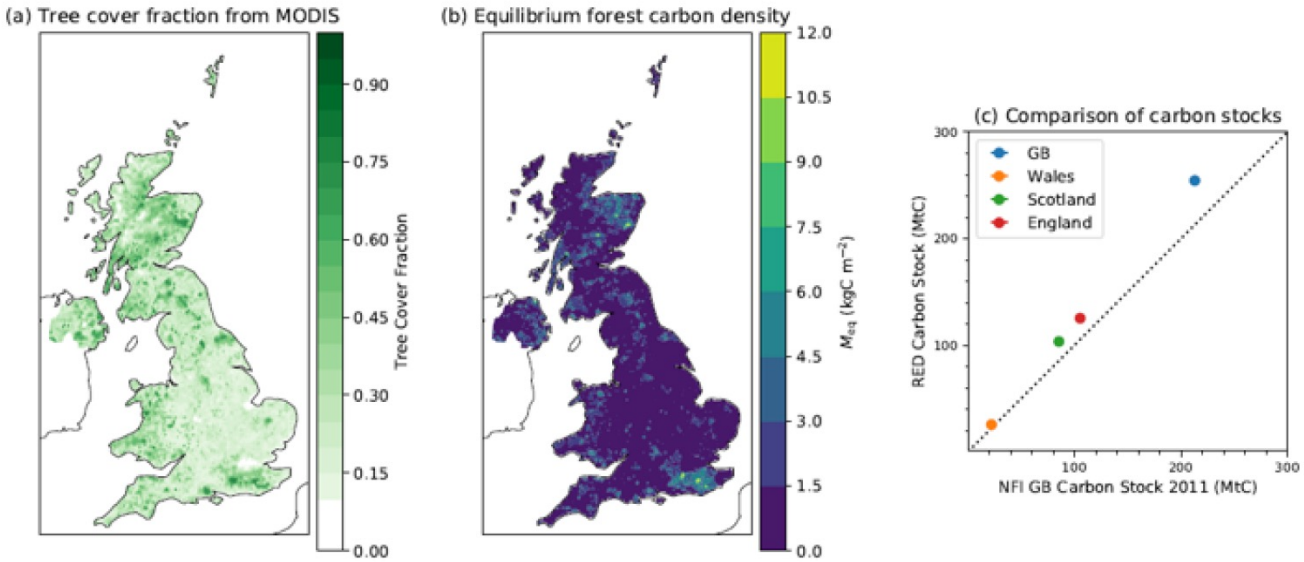
- Which feeds back into other equations such as total carbon density:

$$M = N\left(\frac{3}{4\mu_0}\right)\left(\frac{1}{1+\frac{3}{8\mu_0^2}+\frac{3}{32\mu_0^3}}\right)\left(\frac{3}{8\mu_0^3}+\frac{3}{32\mu_0^4}\right)$$

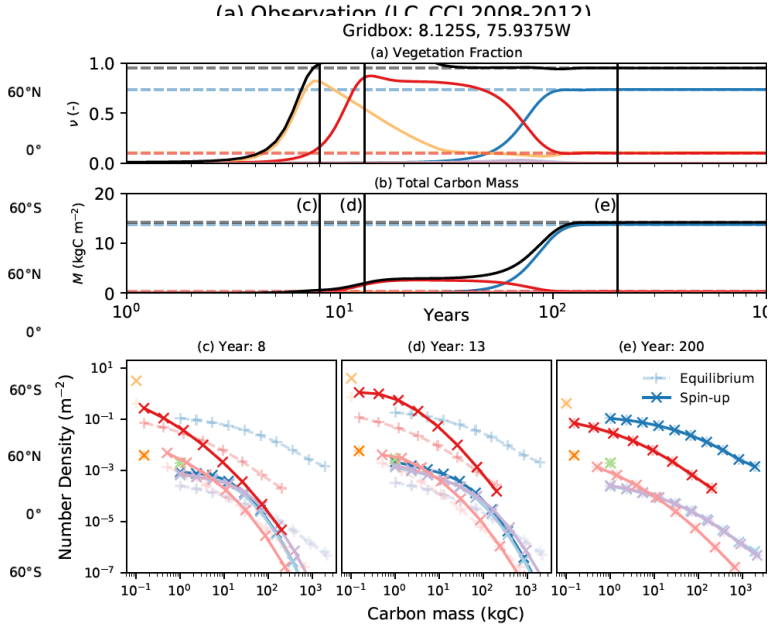
Estimating UK Biomass from MODIS data

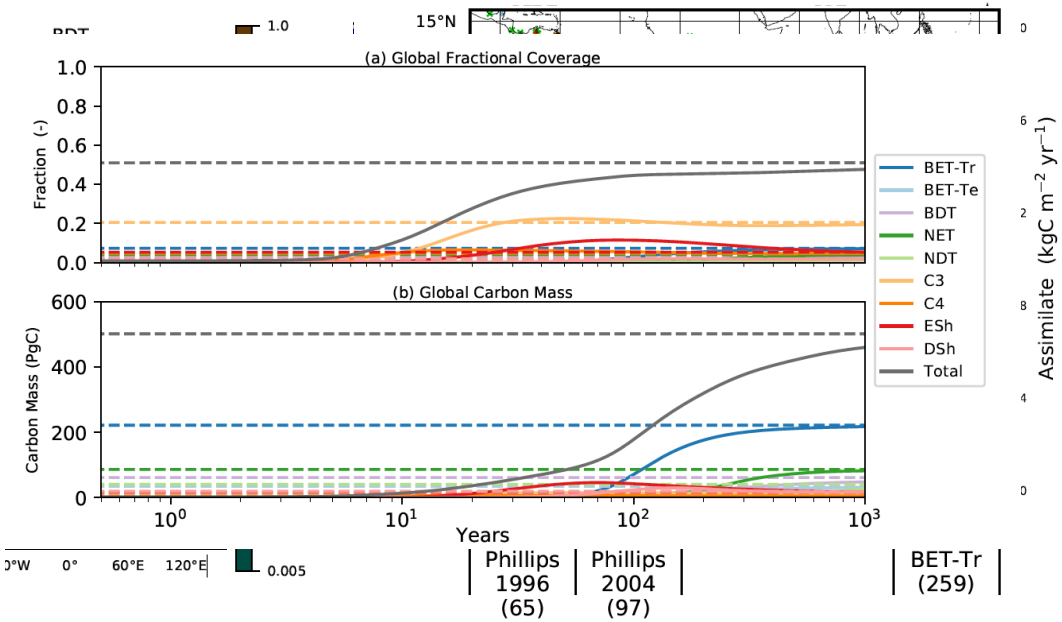
$$\nu = 1 - \left(\frac{1-\alpha}{\alpha}\right)\frac{\mu_0}{1 + \frac{3}{4\mu_0} + \frac{3}{8\mu_0^2} + \frac{3}{32\mu_0^3}}$$

$$M = m_0 N \left(1 + \frac{1}{\mu_0} + \frac{3}{4\mu_0^2} + \frac{3}{8\mu_0^3} + \frac{3}{32\mu_0^4}\right)$$



Finding implicit estimates for mortality or growth rates

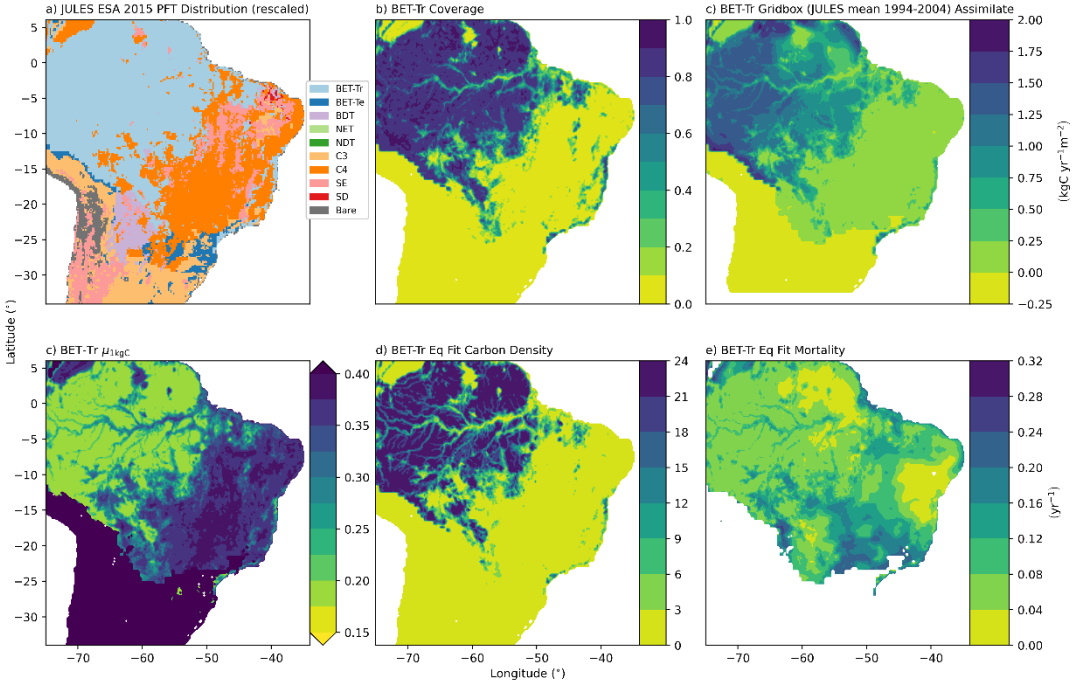




Phillips 1996 (65)	Phillips 2004 (97)	BET-Tr (259)

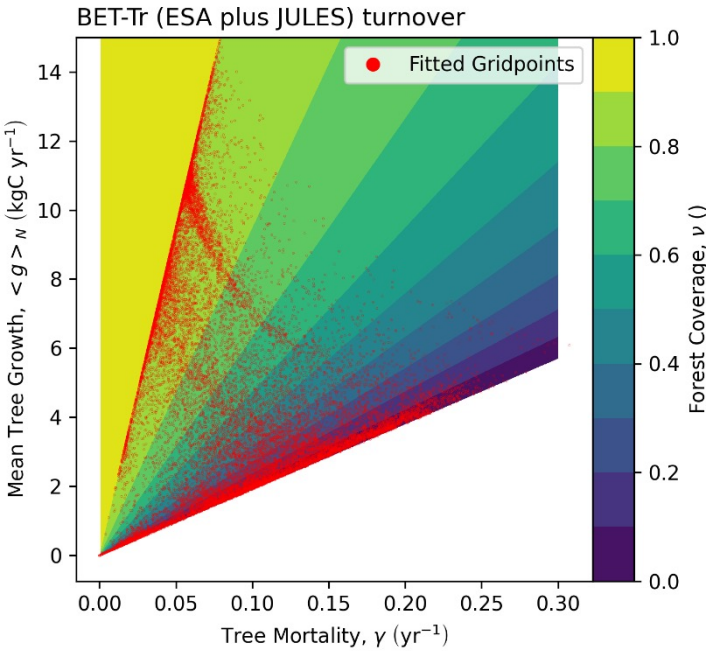
Looking more regionally

- Using higher resolution ESA LC_CCI data proscribed to the 9 JULES PFTs from 2015, we can iterate to find the equilibrium values of μ_0 for the equilibrium ν.
- Coupled with JULES outputs of Net Assimilate from the the factorial experiments for hadGEM2 for Brazil (1994-2004 average).



Conclusions

- RED is a one-dimensional demographic DGVM that focuses on size.
- DET, assumption of metabolic scaling + mortality, seems to describe observed forest size-structure.
- Using RED competition dynamics we can expand DET by solving for recruitment boundary condition.
- This provides us with semi-analytical inversions of observed coverage for the demography and other variables.



References (1)

RED model + DET papers:

Argles, A. P., Moore, J. R., Huntingford, C., Wiltshire, A. J., Harper, A. B., Jones, C. D., & Cox, P. M. (2020). Robust Ecosystem Demography (RED version 1.0): a parsimonious approach to modelling vegetation dynamics in Earth system models. *Geoscientific Model Development*, *13*(9), 4067-4089.

Moore, J. R., Zhu, K., Huntingford, C., & Cox, P. M. (2018). Equilibrium forest demography explains the distribution of tree sizes across North America. *Environmental Research Letters*, *13*(8), 084019.

Moore, J. R., Argles, A. P., Zhu, K., Huntingford, C., & Cox, P. M. (2020). Validation of demographic equilibrium theory against tree-size distributions and biomass density in Amazonia. *Biogeosciences*, *17*(4), 1013-1032

Argles, A. P., Moore, J. R., & Cox, P. M. (2022). Dynamic Global Vegetation Models: Searching for the balance between demographic process representation and computational tractability. *PLOS Climate*, *1*(9), e0000068..

Observations used in LBA evaluation:

Chambers, J., da Silva, R., Tribuzy, E.S., dos Santos, J., & Higuchi, N. (LBA-ECO CD-08 Tree Diameter Measurements, Jacaranda Plots, Manaus, Brazil: 1999-2001.). 2013. ORNL DAAC, Oak Ridge, Tennessee, USA.: Oak Ridge National Laboratory. doi:10.3334/ORNLDAAC/1194

Menton, M., Figueira, M., de Sousa, C., Miller, S., da Rocha, H., & M.L., G. (2011). LBA-ECO CD-04 Biomass Survey, km 83 Tower Site, Tapajos National Forest, Brazil. Oak Ridge National Laboratory Distributed Active Archive Center, Oak Ridge, Tennessee, U.S.A: Oak Ridge National Laboratory. doi:10.3334/ORNLDAAC/990

Saleska, S., da Rocha, H., Huete, A., Nobre, A., Artaxo, P., & Shimabukuro, Y. (2013). LBA-ECO CD-32 flux tower network data compilation, Brazilian Amazon. Oak Ridge, TN, USA: Oak Ridge National Laboratory.

Wofsy, S. C., Saleska, S. R., Pyle, E. H., & Hutyra, L. R. (2008). LBA-ECO CD-10 Tree DBH Measurements at the km 67 Tower Site, Tapajos National Forest. Oak Ridge, Tennessee, U.S.A.: Oak Ridge National Laboratory. doi:10.3334/ORNLDAAC/859

Zanne, A. E., Lopez-Gonzalez, G., Coomes, D. A., Ilic, J., Jansen, S., Lewis, S. L., . . . Chave, J. (2009). Data from: Towards a worldwide wood economics spectrum.

References (2)

ESA Data fitting to JULES 9 PFTs:

Poulter, B., MacBean, N., Hartley, A., Khlystova, I., Arino, O., Betts, R., ... & Peylin, P. (2015). Plant functional type classification for earth system models: results from the European Space Agency's Land Cover Climate Change Initiative. Geoscientific Model Development, 8(7), 2315-2328.

ESA. Land Cover CCI Product User Guide Version 2. Tech. Rep. (2017). Available at: maps.elie.ucl.ac.be/CCI/viewer/download/ESACCI-LC-Ph2-PUGv2_2.0.pdf

Harper, A. B., Cox, P. M., Friedlingstein, P., Wiltshire, A. J., Jones, C. D., Sitch, S., ... & Bodegom, P. V. (2016). Improved representation of plant functional types and physiology in the Joint UK Land Environment Simulator (JULES v4. 2) using plant trait information. Geoscientific Model Development, 9(7), 2415-2440.

Appendix: How to scale to the gridbox (community) level

- We integrate these allometric relationships to work out the sum (or mean, e.g. height) properties for a given PFT community:

$$Y(t) = \int_{m_0}^{\infty} n(m,t) \cdot y_0 \left(\frac{m}{m_0}\right)^{\phi_y} dm$$

- Forest Biomass (M, $\phi_y = 1, y = m$):

$$M(t) = \int_{m_0}^{\infty} n(m,t) \cdot m \, dm$$

- Forest Growth (G, $\phi_y = 3/4, y = g$):

$$G(t) = \int_{m_0}^{\infty} n(m,t) \cdot g_0 \left(\frac{m}{m_0}\right)^{\frac{3}{4}} dm$$

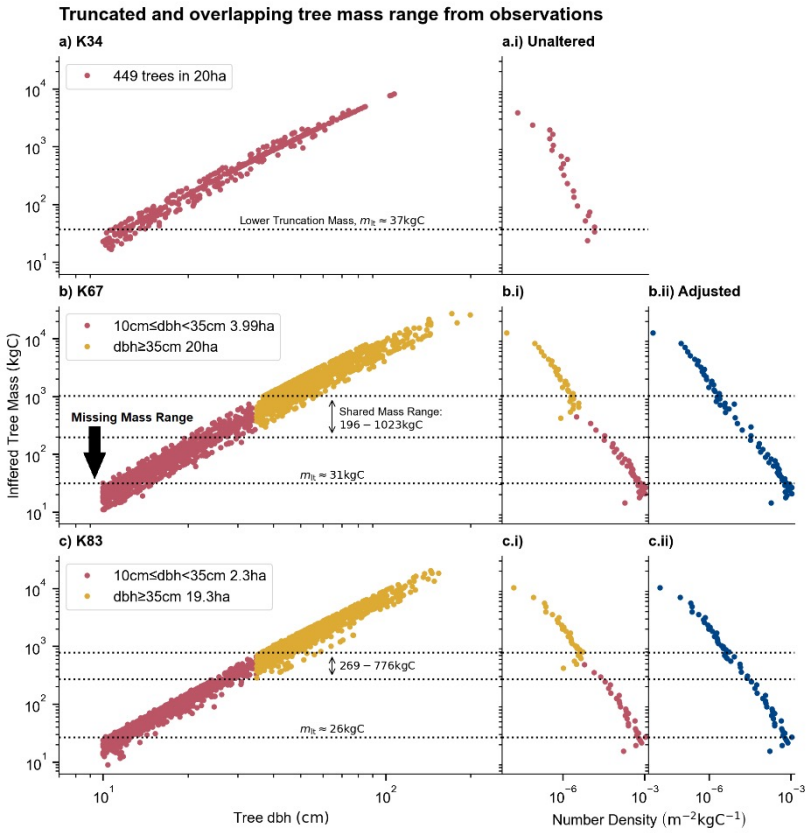
- Forest Coverage (or total projected crown area) (ν, $\phi_y = 1/2$, $y = a$):

$$\nu(t) = \int_{m_0}^{\infty} n(m,t) \cdot a_0 \left(\frac{m}{m_0}\right)^{\frac{1}{2}} dm$$

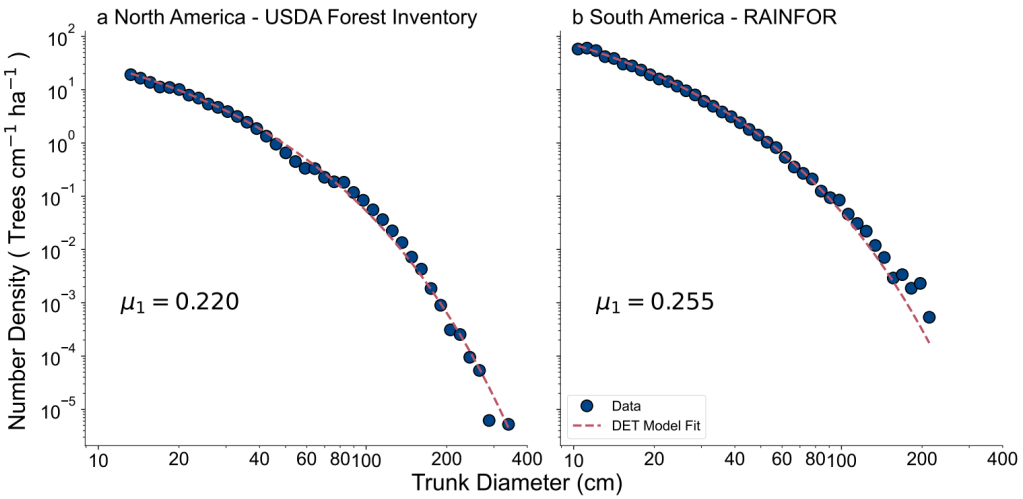
Appendix: LBA Observations

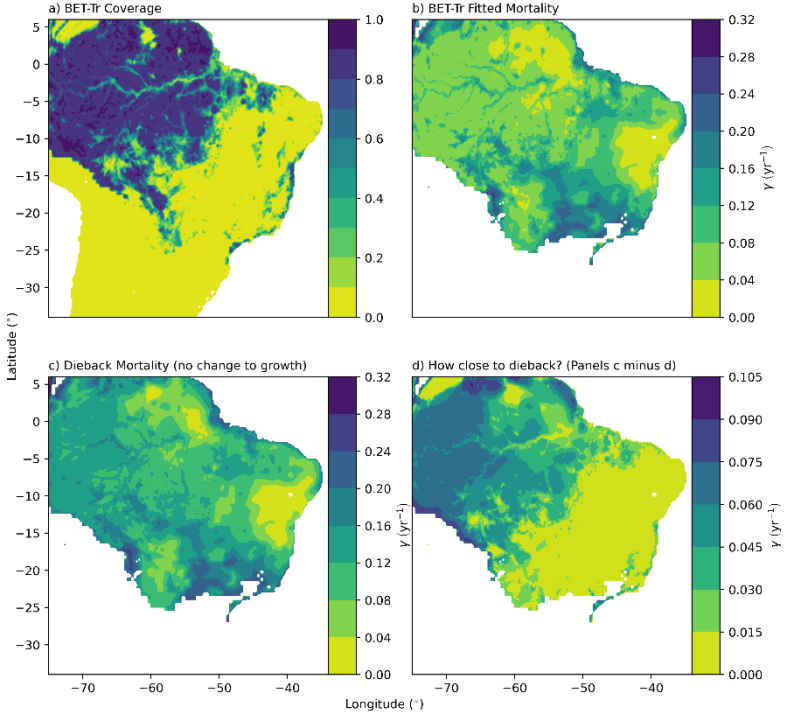
- In addition to previous experience on the suite. Some of these sites (Through LBA-ECO) have small scale species, wood density (also found through the DRYAD database) and dbh data.
- These observations give us a way of getting number density if we divide through by the area of the study.

Sources: Wofsy, et al., 2008 (K34), Menton, et al., 2011 (K83), Chambers, et al., 2013 (K67), and Zanne, A. E., 2009 (DRYAD).



Appendix: more DET fits, but to dbh



Appendix: Forest resilience to increased mortality

- Assuming no change in growth (which ignores the effects of CO2 fertilisation and drought), how much more do we have to increase the background mortality to see complete dieback for BET-Tr with RED?
- Results suggest that the west of the Amazon is more resilient than the east.